

Polar Area

Section 10.

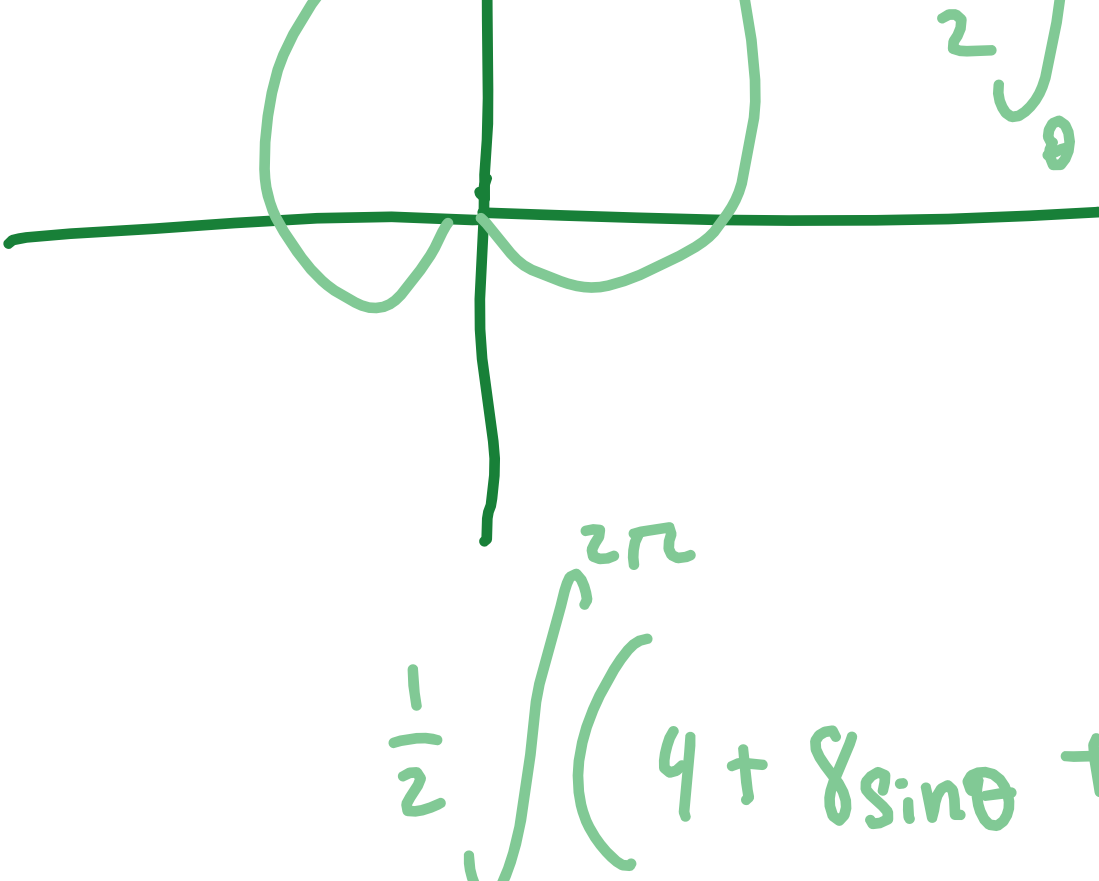
04-02-26

$$\frac{\text{Area of Sector}}{\pi r^2} = \frac{\Theta}{2\pi}$$

$$A = \frac{\Theta}{2\pi} \cdot \pi r^2$$

$$= \frac{1}{2} \int_{\theta_1}^{\theta_2} r^2 d\theta$$

Ex: $r = 2 + 2\sin(\theta)$



$$\frac{1}{2} \int_0^{2\pi} (2 + 2\sin\theta)^2 d\theta$$

$$\frac{1}{2} \int_0^{2\pi} (4 + 8\sin\theta + 4\sin^2\theta) d\theta$$

$$\frac{1}{2} \int_0^{2\pi} 4 + 8\sin\theta + 4\left(\frac{1}{2}\right)(1 - \cos(2\theta)) d\theta$$

$$\int_0^{2\pi} (2 + 4\sin\theta + 1 - \cos(2\theta)) d\theta$$

$$3\theta - 4\cos\theta - \frac{1}{2}\sin(2\theta) \Big|_0^{2\pi}$$

$$(6\pi - 4) - (-4) = 6\pi$$

Find area of one petal in $r = 2\sin(3\theta)$



$$0 = 2\sin(3\theta)$$

$$3\theta = 0, \pi$$

$$\theta = 0, \pi/3$$

$$\frac{1}{2} \int_0^{\pi/3} (2\sin(3\theta))^2 d\theta$$

Area of one petal of $r = 4\cos(2\theta)$



$$0 = 4\cos(2\theta)$$

$$\frac{\pi}{2} = 2\theta$$

$$\theta = \pi/4$$

$$2 \cdot \frac{1}{2} \int_0^{\pi/4} (4\cos(2\theta))^2 d\theta$$

$$\int_0^{\pi/4} 16\cos^2(2\theta) d\theta$$

$$8 \int_0^{\pi/4} (1 + \cos(4\theta)) d\theta$$

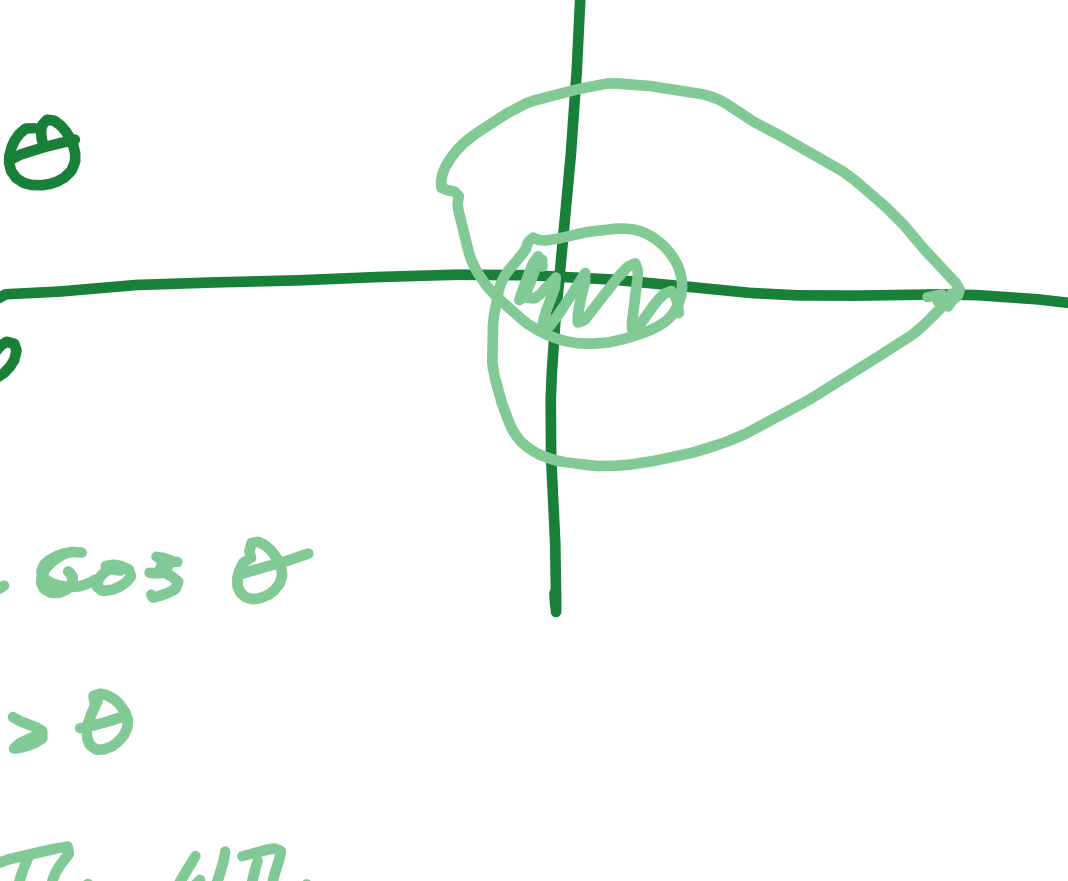
$$8\left(\theta + \frac{1}{4}\sin(4\theta)\right) \Big|_0^{\pi/4}$$

$$8\left(\frac{\pi}{4}\right) = 2\pi$$

#6 on Polar WS 2

$$r = 1 + 2\cos\theta$$

area of inner loop



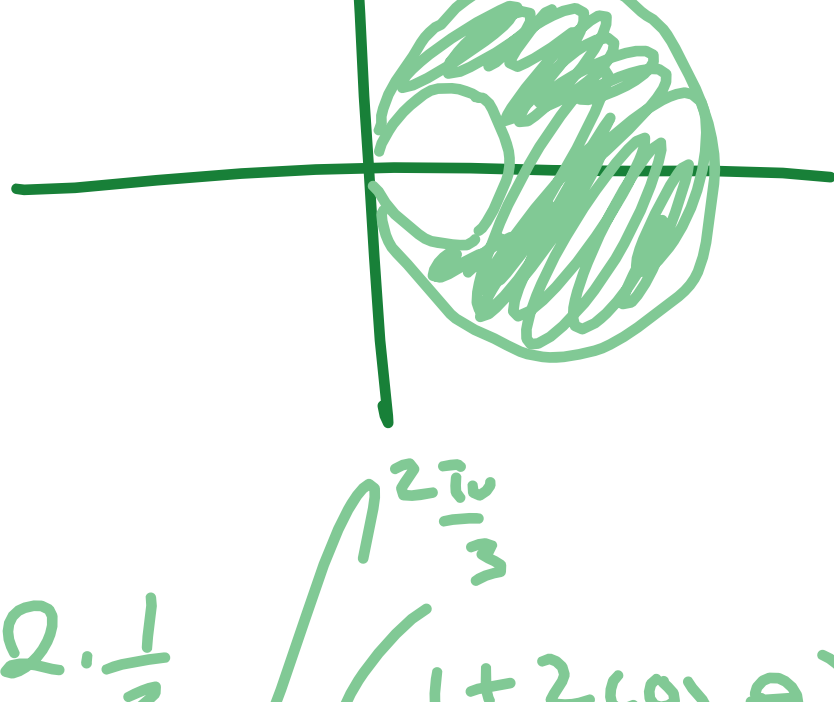
$$0 = 1 + 2\cos\theta$$

$$-\frac{1}{2} = \cos\theta$$

$$\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\int_{\frac{2\pi}{3}}^{\frac{4\pi}{3}} (1 + 2\cos\theta)^2 d\theta$$

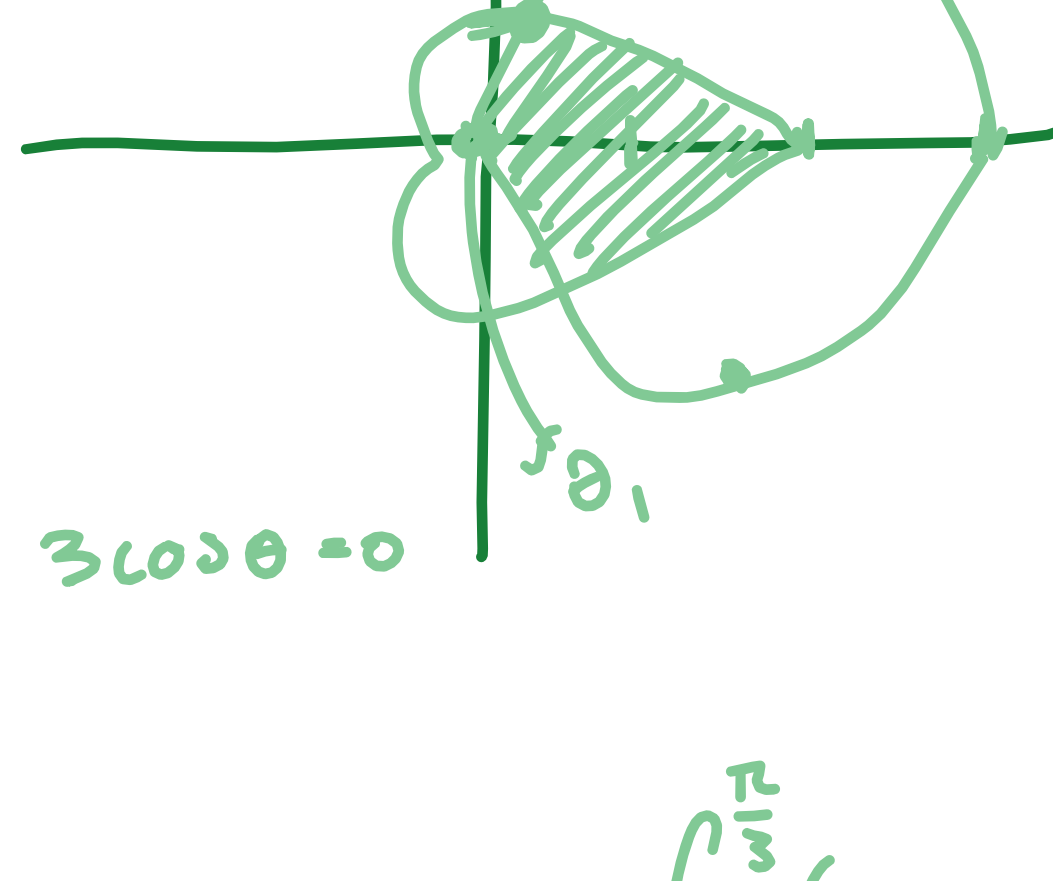
between loops of $r = 1 + 2\cos\theta$



$$2 \cdot \frac{1}{2} \int_0^{\frac{2\pi}{3}} (1 + 2\cos\theta)^2 d\theta$$

Common Interval of $r = 3\cos\theta$

$$r = 1 + \cos\theta$$



$$3\cos\theta = 1 + \cos\theta$$

$$2\cos\theta = 1$$

$$\cos\theta = \frac{1}{2}$$

$$\frac{\pi}{3}, \frac{5\pi}{3}$$

$$\frac{1}{2} \int_{-\pi/3}^{\pi/3} (1 + \cos\theta)^2 d\theta$$

$$\int_0^{\pi/3} (1 + \cos\theta)^2 d\theta$$

$$2 \cdot \frac{1}{2} \int_{\pi/3}^{\pi/2} (3\cos\theta)^2 d\theta + \int_0^{\pi/3} (1 + \cos\theta)^2 d\theta$$